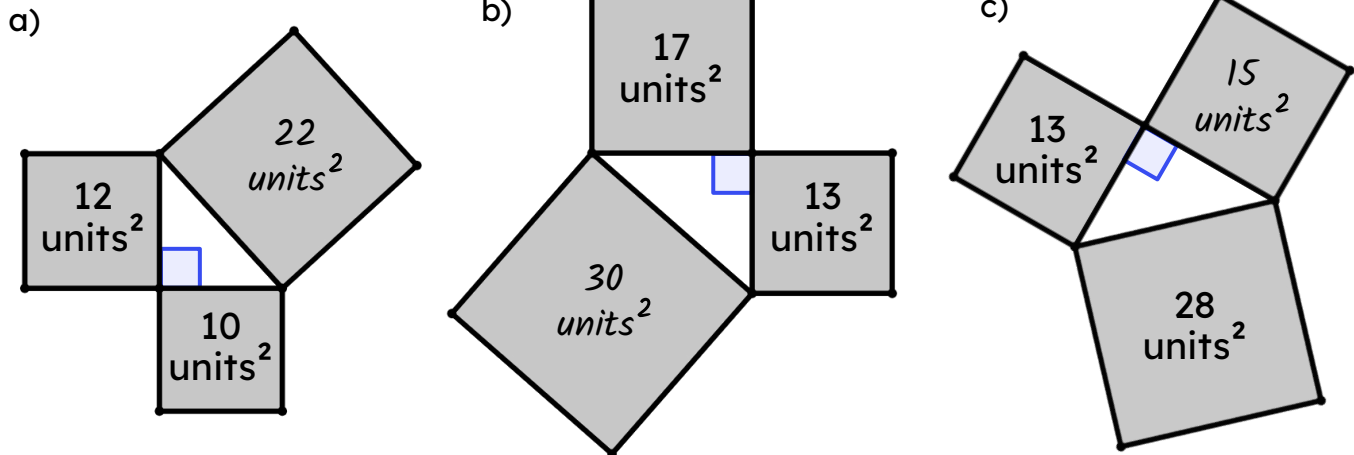




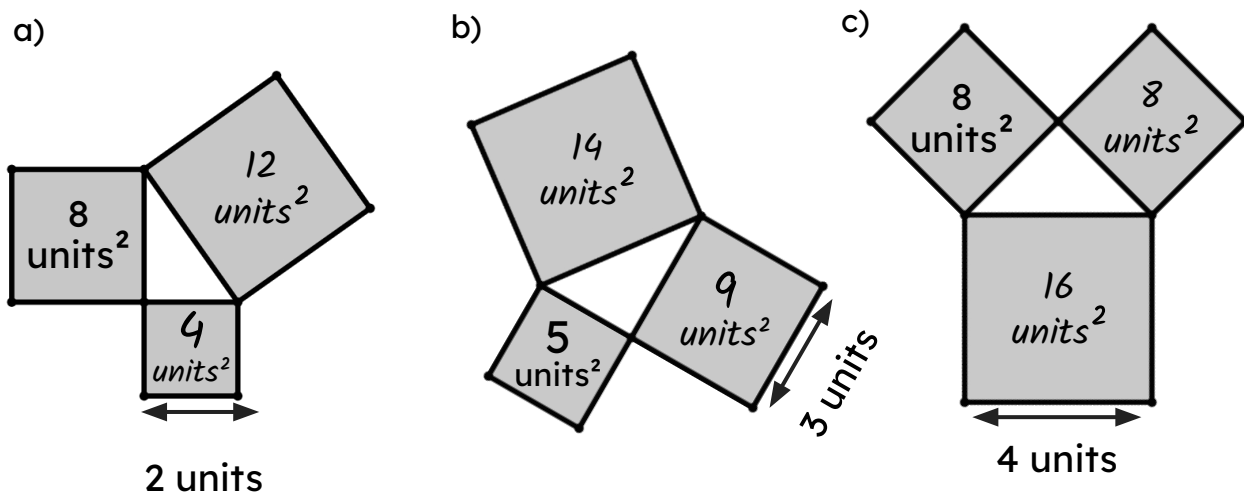
Further demonstrating of Pythagoras' theorem

Task A: Generating right-angles triangles from squares

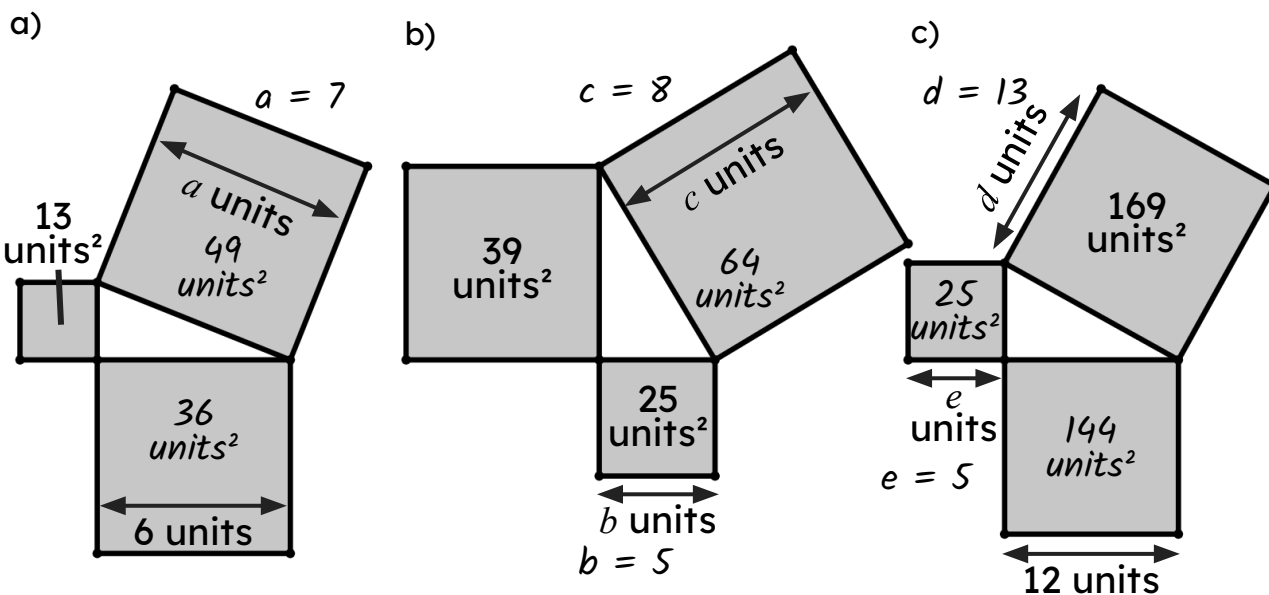
1) For each of these right-angled triangles, find the area of each unlabelled square.



2) For each of these right-angled triangles, find the area of all squares.

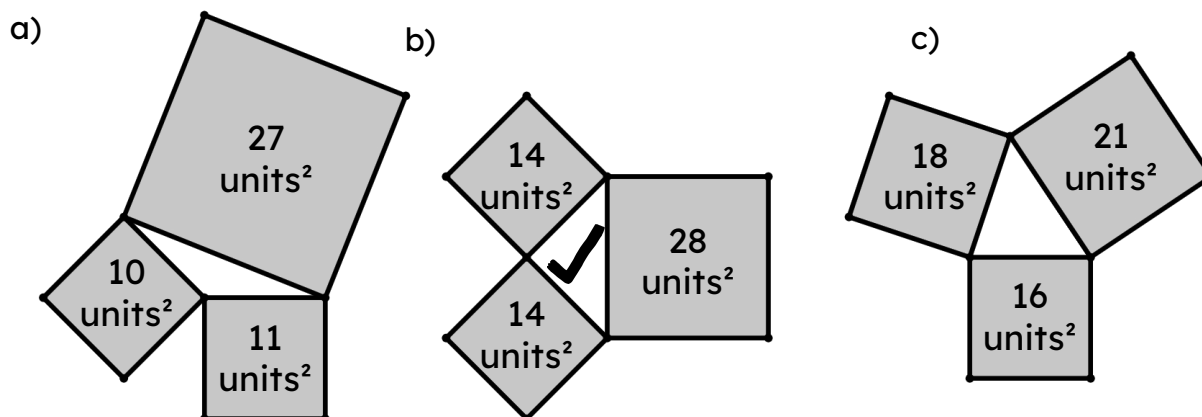


3) For each of these right-angled triangles, fill in the area of each square and find the missing length of each square that is labelled with a letter.



Task B: Introducing Pythagoras' theorem

1) On which of these diagrams is **Pythagoras' theorem** true?



2) Explain your reasoning for question 1).

b is the only diagram where the sum of the two smaller areas is equal to the larger area. This is the only right-angled triangle.

3) Which of these are valid representations of **Pythagoras' theorem** for these three squares which enclose a right-angled triangle?

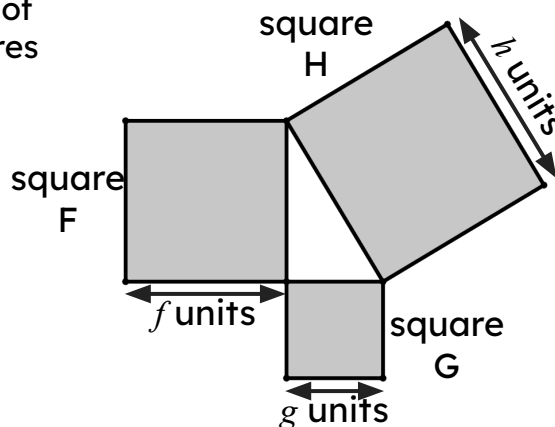
area of square F plus the area of square G equals the area of

square H

a $f^2 + g^2 = h^2$ ✓

b $f^2 = g^2 + h^2$

c $h^2 = g^2 + f^2$ ✓



4) a) What could 15^2 represent in this diagram?

15^2 is the area of square B

b) What is the area of the largest square?

Square C has an area of 289 units²

c) Fill in the blanks for these calculations of **Pythagoras' theorem**.

$$8^2 + 15^2 = 17^2$$

$$64 + 225 = 289$$

area of
square
A

area of
square
B

area of
square
C

